

KALUMBURU, Australia

WMO: 941000

Lat: 14.2964S

Lon: 126.6453E

Elev: 79

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	49.9	52.7	23.9	18.1	78.7	27.6	21.6	80.0	19.5	84.8	15.4	83.0	1.6	200	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	19.1	99.9	74.8	98.3	74.7	96.7	75.0	82.5	92.1	81.1	91.2	80.1	90.3	9.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	156.3	87.7	78.6	148.7	86.9	77.4	143.0	85.4	46.4	92.0	44.9	91.4	43.8	90.5	98.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.8	14.8	13.3	DB	45.3	102.4	3.1	2.2	43.1	104.0	41.2	105.3	39.5	106.5	37.2	108.1
			WB	42.7	83.6	2.6	1.1	40.8	84.4	39.3	85.0	37.9	85.7	36.0	86.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.9	84.1	83.4	83.4	81.9	77.6	73.3	73.5	74.9	80.5	85.0	86.9	86.2	(6)
	DBStd	5.76	2.81	2.65	2.45	2.74	4.19	4.94	4.06	3.27	3.74	2.60	2.27	3.11	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	5	0	0	0	0	0	4	1	0	0	0	0	0	(9)
	CDD50	11270	1056	934	1037	958	855	700	728	773	916	1084	1107	1122	(10)
	CDD65	5800	591	514	572	508	390	254	264	308	466	619	657	657	(11)
	CDH74	67430	6655	5404	5987	5382	4349	2988	3252	3740	5541	7615	8444	8073	(12)
	CDH80	33080	2862	2188	2687	2802	2199	1332	1497	1797	2977	4139	4532	4068	(13)
(14) Wind	WSAvg	4.9	4.9	4.4	4.0	4.1	4.8	5.0	4.8	5.1	5.5	5.9	5.4	5.0	(14)
(15) Precipitation	PrecAvg	49.8	12.8	12.2	8.6	2.6	0.6	0.0	0.2	0.0	0.1	1.1	2.6	7.7	(15)
	PrecMax	90.1	31.9	31.9	23.9	14.2	4.1	0.6	4.8	0.2	1.7	6.1	7.6	22.6	(16)
	PrecMin	28.5	2.3	1.2	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	1.5	(17)
	PrecStd	12.5	6.5	5.9	5.3	3.1	1.1	0.1	0.8	0.0	0.3	1.2	1.7	5.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.5	97.7	98.1	98.7	96.2	94.3	94.1	94.6	99.5	101.9	101.5	102.7	(19)
		MCWB	79.8	78.0	77.4	72.5	70.6	69.8	65.8	64.7	68.9	71.5	75.0	77.2	(20)
	2%	DB	95.7	94.9	95.6	96.2	94.3	91.7	92.1	92.7	97.0	99.0	99.3	99.6	(21)
		MCWB	79.5	79.3	78.2	73.5	69.9	67.9	65.1	64.3	69.2	72.6	75.4	77.9	(22)
	5%	DB	93.6	92.9	93.9	94.7	92.6	89.8	90.2	91.0	94.9	96.9	97.3	96.6	(23)
		MCWB	79.0	78.8	78.2	73.8	69.6	66.3	64.8	64.7	69.7	73.2	75.9	78.1	(24)
	10%	DB	91.4	90.7	91.8	92.8	90.7	87.7	88.2	89.0	92.8	94.9	95.3	94.2	(25)
		MCWB	78.5	78.6	78.1	73.7	68.9	65.3	64.1	64.7	69.6	73.8	76.1	77.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.4	83.4	83.6	82.4	78.8	75.1	73.0	74.0	83.5	82.0	82.1	82.7	(27)
		MCDB	93.6	91.5	91.9	91.6	86.5	85.7	83.8	85.8	92.2	94.2	91.5	94.4	(28)
	2%	WB	81.2	81.5	81.5	80.1	76.3	73.0	71.0	72.4	76.3	78.9	80.1	80.8	(29)
		MCDB	91.5	90.3	91.4	90.7	86.2	84.2	83.5	84.7	88.4	91.8	90.9	92.1	(30)
	5%	WB	80.0	80.2	80.1	78.4	74.8	71.1	69.2	71.0	74.8	77.4	79.0	79.8	(31)
		MCDB	90.0	89.2	90.3	89.3	84.3	83.2	81.8	83.0	86.8	89.2	89.7	90.8	(32)
	10%	WB	79.1	79.2	79.0	77.1	73.4	68.9	67.5	69.5	73.7	76.5	78.1	79.0	(33)
		MCDB	88.5	87.9	88.8	87.1	82.5	82.2	81.0	81.6	85.5	87.6	88.7	89.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	14.3	17.2	22.8	27.0	29.2	30.8	30.5	26.8	22.5	19.1	17.1	(35)
		MCDBR	18.3	18.6	21.0	25.8	28.2	29.7	32.0	32.4	28.1	25.0	21.5	20.6	(36)
	5% WB	MCWBR	6.3	7.4	7.9	8.9	9.0	10.0	10.5	10.3	7.9	6.5	6.0	6.2	(37)
		MCWBR	16.5	16.8	18.5	20.4	21.3	25.9	27.5	26.1	23.7	20.8	19.1	18.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.423	0.414	0.396	0.384	0.372	0.346	0.344	0.359	0.417	0.446	0.447	0.424	(40)
		taud	2.464	2.493	2.540	2.522	2.503	2.535	2.523	2.470	2.354	2.302	2.341	2.446	(41)
	Ebn at Noon		292	293	292	285	278	280	284	288	281	280	284	292	(42)
		Edh at Noon	38	36	34	33	31	29	30	34	40	44	43	38	(43)
(44) All-Sky Solar Radiation	RadAvg		1811	1840	1851	1838	1706	1630	1720	1945	2142	2213	2188	1954	(44)
	RadStd		202	269	226	152	104	69	58	48	69	98	85	164	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page